SECAI+ LESSON 6.2.1

Principles of Responsible AI

What is AI risk? Who is accountable? How do we build AI that is fair, safe, and trustworthy?

- AI Risk
- Core Principles
- Human Oversight
- Inclusiveness

Understanding AI Risk

AI risk is **technical** (wrong outputs, model errors) or **non-technical** (reputational damage, unexpected behavior).

1

Business Context

What decision does AI influence? What if it makes a poor recommendation?

2

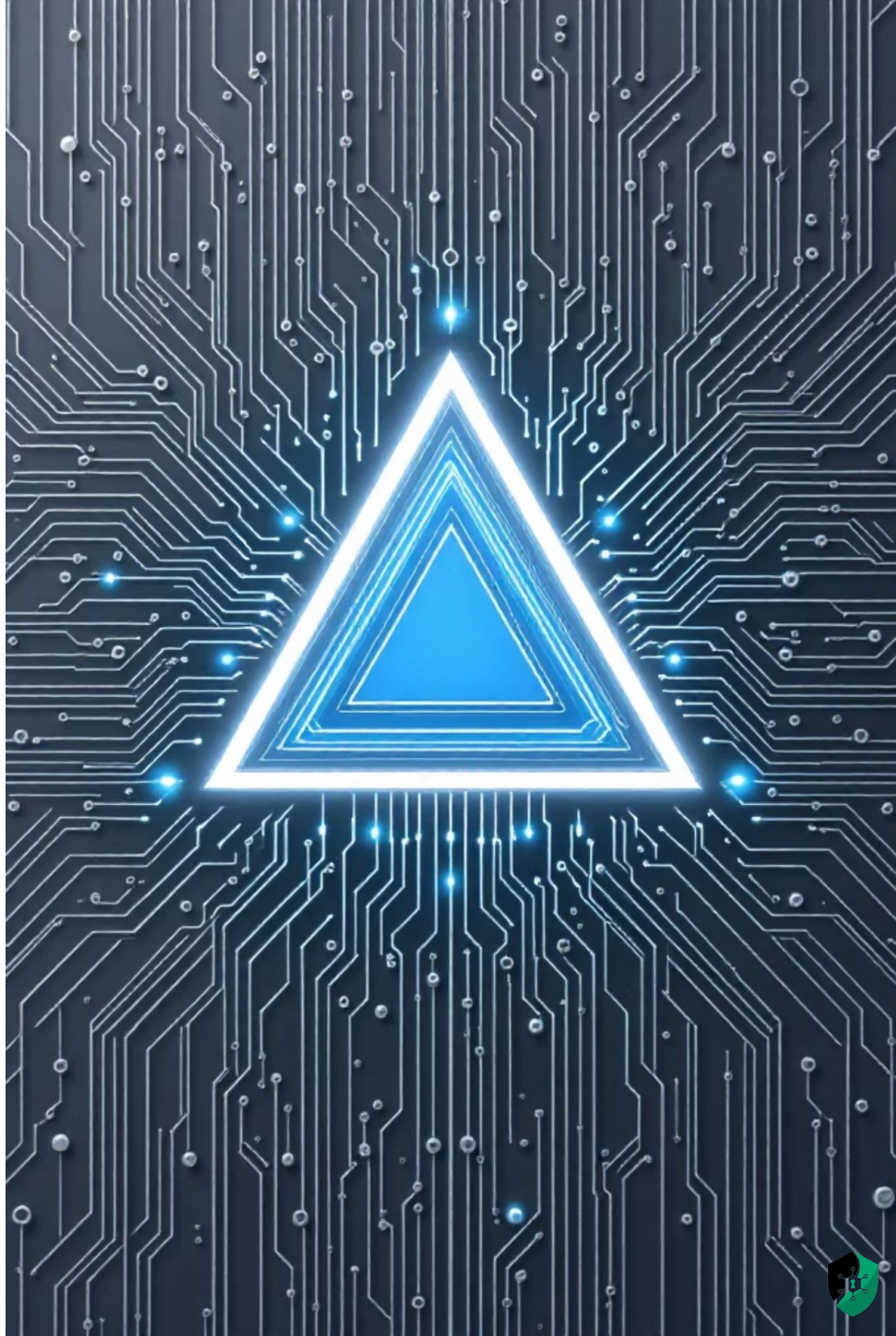
Data & Model Behavior

How does the system learn? Can patterns become unreliable or skewed?

3

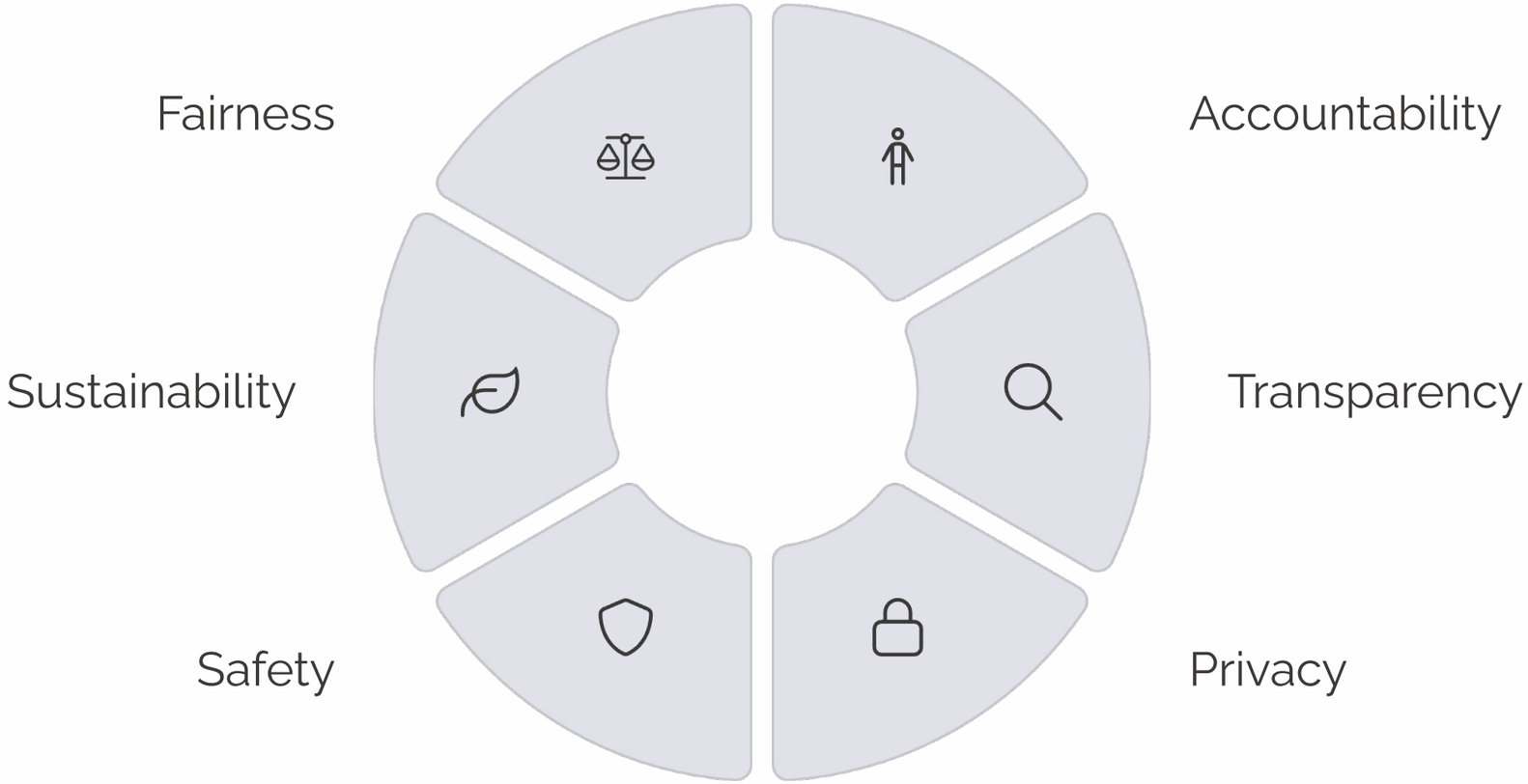
Operations & Oversight

Deployment, monitoring, incident handling, and accountability.



Principles of Responsible AI

Broad values translated into **specific design and control expectations**.



Fairness

Outcomes must not disadvantage individuals based on protected or irrelevant characteristics.

In Practice

- Examine training data for **historical imbalances**
- Measure results **across different groups**
- Establish clear rules for **acceptable disparities**

Accountability & Transparency

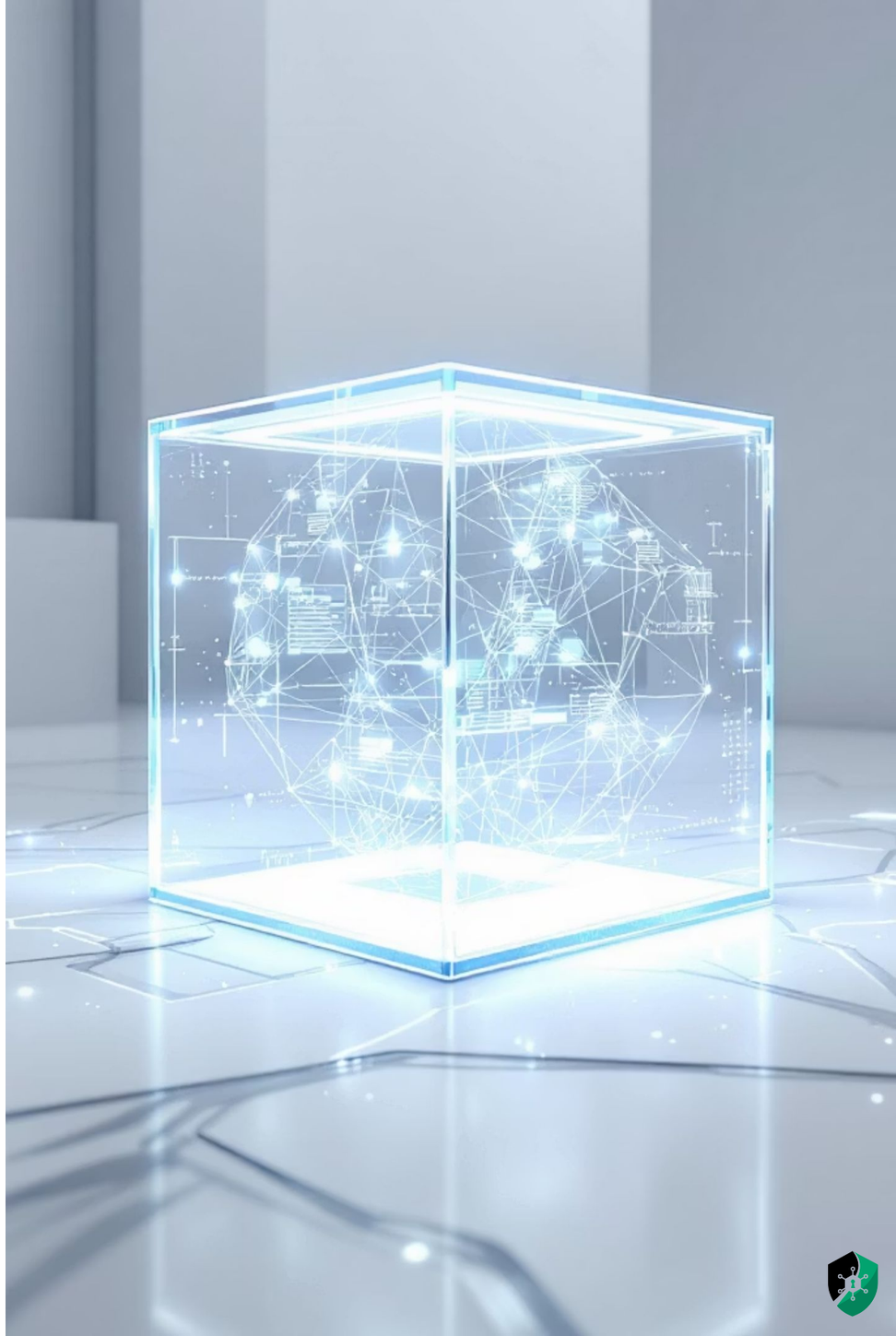
Accountability

Clear ownership of AI decisions ensures **quicker resolution** when model behavior changes.

> Who is responsible when the AI gets it wrong?

Transparency & Explainability

- Document models and data
- Maintain decision logs
- Summarize key factors driving each prediction



Privacy & Data Governance

Collect and use data in a **legitimate, minimal, and well-protected** manner.

Consent Management

Users must know how their data is used.

Retention Limits

Don't keep data longer than necessary.

De-identification

Remove or mask personal identifiers where possible.

Prompt Filtering

Screen inputs and outputs for personal data leakage.

Safety & Security

Prevent harm from **misuse and attacks**.

Adversarial Manipulation

Inputs crafted to fool or destabilize the model.

Data Poisoning

Corrupting training data to skew model behavior.

Prompt Injection

Malicious instructions hidden in user input.

Insecure APIs & Misuse

Using AI tools (e.g. code generation) to create harmful scripts.

Human Oversight

Human-in-the-Loop

AI proposes → Human decides.

Every significant action requires explicit human approval before execution.

Human-on-the-Loop

AI decides routine cases → Humans monitor & can override.

For **high-stakes decisions**, human review or strict validation is always essential.

The higher the stakes, the more human involvement is required.

Sustainability, Consistency & Inclusiveness

Sustainability

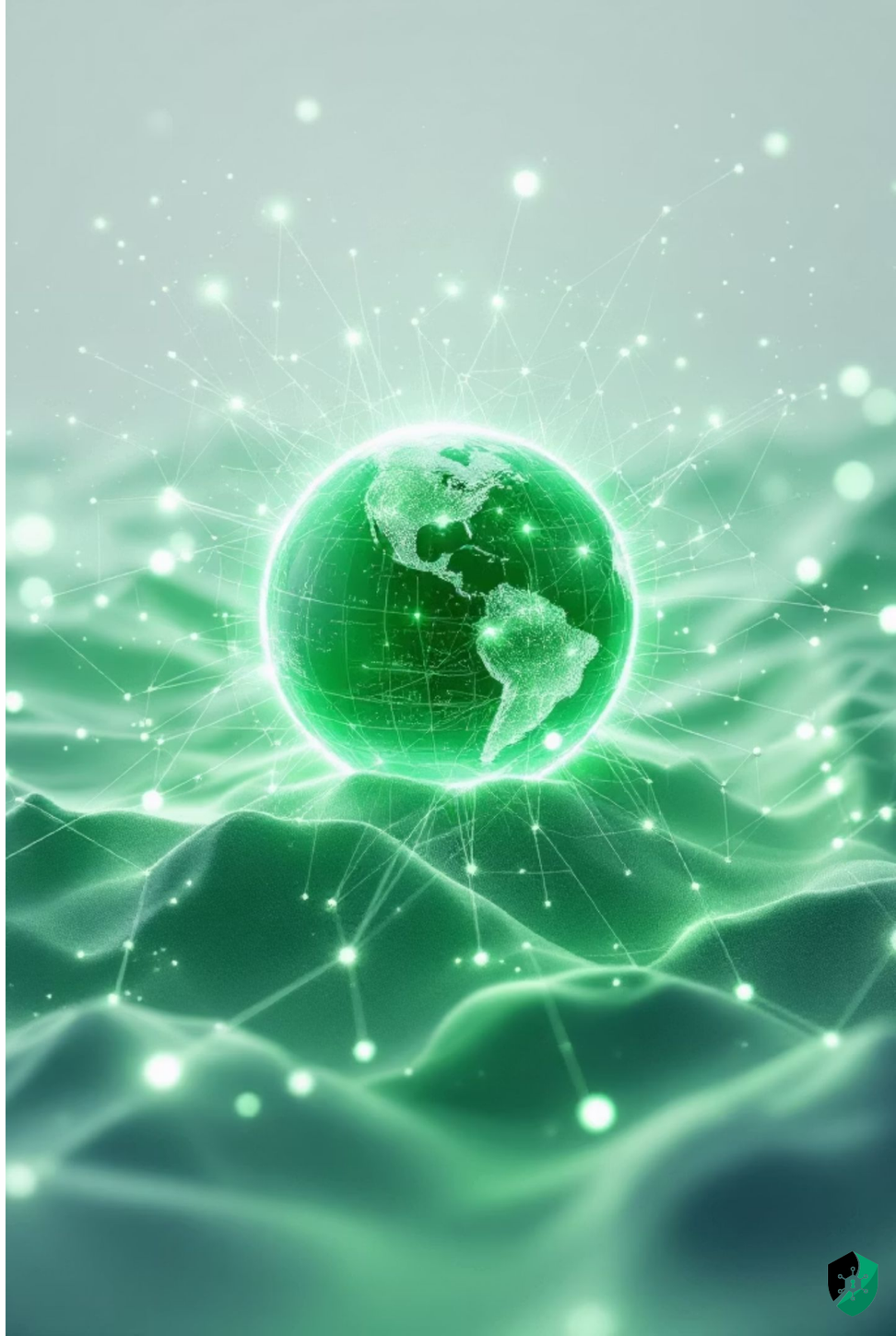
Monitor energy and resource costs of training and operating large models. Seek more efficient alternatives.

Consistency

Stable, equitable results across all demographics. Conduct bias audits using **demographic parity** and **equal opportunity** metrics.

Inclusiveness

Design AI to serve **diverse populations** equitably — no group should be left behind.



Key Takeaways

1 **AI risk is multi-dimensional**

Technical errors, model bias, and operational failures all contribute to risk across business, data, and oversight lines.

2 **Six core principles guide responsible AI**

Fairness, Accountability, Transparency, Privacy, Safety, and Sustainability convert values into actionable controls.

3 **Human oversight is non-negotiable**

High-stakes AI decisions always require human review — either in-the-loop or on-the-loop.

4 **Bias audits and inclusiveness are ongoing**

AI systems must be continuously tested for fairness across all populations, not just at launch.